B. E– 3rd Year V- Sem- Course-Based-Project (2026-27)

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ABSTRACT

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Title of the Project:

Landrover Robot Operated by Cell Phone

ABSTRACT:

This project presents the design and development of a **cellphone-controlled Land Rover robot** capable of wireless movement and navigation using commands provided through a mobile phone. The proposed system is designed to provide a simple and convenient method of controlling a robotic vehicle remotely without the need for direct physical interaction. The robot consists of a microcontroller, motor driver, DC geared motors, wheels, and a wireless communication module for receiving commands from the cellphone. The mobile phone is used to send directional commands such as forward, backward, left, right, and stop to the robot. The microcontroller processes the received commands and controls the motors accordingly through the motor driver circuit. The system is assembled and tested to verify the accuracy of movement, wireless communication, and response to different control commands. The project demonstrates the application of embedded systems, wireless communication, and motor control in developing a simple remotely operated mobile robotic platform.

Keywords:
Land Rover Robot, Cellphone Control, Wireless Communication, Microcontroller, Motor Driver, DC Geared Motor, Robotics, Embedded Systems, Remote Control.

Methodology:

- Design and assemble the mechanical structure of the Land Rover robot.
- Interface the microcontroller with the motor driver and DC geared motors.
- Establish wireless communication between the cellphone and the robot.
- Develop a cellphone-based control interface for sending movement commands.
- Program the microcontroller to receive and interpret commands from the cellphone.
- Control the motors according to the received commands for forward, backward, left, and right movement.
- Implement a stop command to halt the robot whenever required.
- Test the robot under different operating conditions and control commands.
- Evaluate the response time, movement accuracy, wireless communication range, and overall performance.
- Optimize the control system and verify the reliable operation of the cellphone-controlled Land Rover robot.

Tools:

Hardware: Arduino UNO, Motor Driver, DC Geared Motors, Wheels, Chassis, Wireless Communication Module, Battery, Jumper Wires.

Software: Arduino IDE, Mobile Phone/Control Application.

References:

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K. Akul Kumar

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